

Matrix-Dilated Fabius–Rvachev Laws

Infinite Box Splines, Thue–Morse Directional Derivatives, Bell–Bernoulli Tensor Resolvents, and Rotating-Zonoid q -Series

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Repository-aware proofs and reproducible numerical experiments

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Abstract

The current ProveIt corpus develops the one-dimensional Fabius–Rvachev system unusually far: exact dyadic arithmetic, Thue–Morse derivative trees, q -binomial interpolation, infinite sinc products, Bell–Bernoulli moments, Legendre and Appell representations, inverse-function jets, and phase-aware Lambert– W endpoint asymptotics are all already present. This report moves in a direction not found in the recursively audited LaTeX tree: an expansive matrix replaces scalar dilation, and a finite vector dictionary replaces the single uniform digit.

Let $A \in \text{GL}_d(\mathbb{R})$ be expansive, let $V = [v_1 \ \cdots \ v_r]$, and let the vectors U_n have independent $\text{Unif}[-1, 1]$ coordinates. We study

$$X_{A,V} = \sum_{n \geq 1} A^{-n} V U_n.$$

Its support is the self-affine infinite zonotope $K_{A,V} = \sum_{n,j} [-A^{-n}v_j, A^{-n}v_j]$, while its Fourier transform is the matrix-Mahler product

$$\Phi_{A,V}(\xi) = \prod_{n \geq 1} \prod_{j=1}^r \text{sinc}_{\text{rad}}(\langle \xi, A^{-n}v_j \rangle), \quad \Phi_{A,V}(A^T \xi) = \prod_j \text{sinc}_{\text{rad}}(\langle \xi, v_j \rangle) \Phi_{A,V}(\xi).$$

A finite-orbit spanning condition is shown to be exactly the condition for a full-dimensional compactly supported C^∞ density. The proof supplies a log-Gaussian Fourier bound. The limiting density is log-concave, while a simple matrix-geometric criterion detects when it nevertheless has an exact central plateau. Finite prefixes are centered box splines and converge in every C^m norm at a geometric rate; their first corrections are explicit Bell–Bernoulli differential operators.

The scalar Thue–Morse derivative formula lifts to an exact multivariate cube. After differentiating once along every generator in the first N scales, the density becomes a signed sum of 2^{rN} affine copies of itself, with signs $(-1)^{\text{wt}_2(k)}$. The same cube gives a tensor Prouhet identity annihilating every polynomial of total degree below rN . All joint cumulants are obtained from the resolvent

$$\kappa_{2m}(X_{A,V}) = \frac{2^{2m-1} B_{2m}}{m} (A^{\otimes 2m} - I)^{-1} \sum_{j=1}^r v_j^{\otimes 2m},$$

and multivariate Bell partitions then give every moment. This yields rationality and denominator bounds for rational data, an exact Appell–Koopman intertwining law, and explicit Legendre coefficients for every one-dimensional projection.

The planar family $A^{-1} = qR_\theta$ produces the most concrete new geometry. Its support has perimeter $4\|v\|q/(1-q)$ and area

$$\frac{4\|v\|^2 q^2}{1-q^2} \sum_{m \geq 1} |\sin(m\theta)| q^m.$$

For rational θ/π this area series is rational in q ; for irrational θ/π the unit circle is its natural boundary. Irrational supports have a dense set of exposed edges whose union has full boundary length. Nevertheless, a natural Abel rescaling makes them converge to a Euclidean disk as $q \uparrow 1$; rational rotations instead converge to finite zonotopes. Exact directional-cap Laplace products connect this geometry back to inverse Fabius and Lambert– W questions. The irrational endpoint modulation is isolated as a conjectural rotation-cocycle problem. A fully commented Python program reproduces all figures and numerical checks.

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1 Corpus audit, contribution boundary, and notation

1.1 Recursive audit

The report is pinned to commit 90b2d3587833dea1bc4a6bf0c36d92f7200396e6 of <https://github.com/VladimirReshetnikov/ProveIt>. A recursive GitHub code inventory returned 119 files with extension `.tex` under `Analysis/FabiusFunction/docs`. The tree contains current expositions, large consolidated frontier volumes, same-day incoming reports, paper transcriptions, and frozen historical copies. Reading every frozen predecessor as if it were an independent claim would overcount the corpus. The audit therefore used the repository’s own `MANIFEST.md`, coverage tables, and provenance notes to identify canonical descendants, and then searched the whole recursive tree for exact and close variants of every proposed definition and formula.

The most important current descendants inspected directly were the primary Fabius–Rvachev exposition, the representation atlas, the exponent/ q -series volume, the inverse-and-sampling volume, the spectral-arithmetic volume, the Thue–Morse atlas, the consolidated frontier reports, the mathematical glossary, the Lean walkthrough, and the newest information-geometry report. Targeted whole-tree collision searches included

Concept family	Search phrases and close variants
matrix geometry	matrix dilation, self-affine, zonotope, zonoid, support function
multivariate splines	multivariate box spline, infinite box spline, matrix sinc product
tensor algebra	tensor cumulant, tensor Bell, Lyapunov cumulant, matrix Appell
rotating q -series	rotating support, $ \sin(n\theta) $, natural boundary, Abel shape
directional endpoints	cap distribution, support deficit, rotation cocycle

No occurrence of the defining matrix-dilated family, the rotating-zonoid area series, its natural-boundary theorem, or the tensor Prouhet cube was found. The word “new” below consequently means *new to this audited repository snapshot*, not a claim of universal publication priority.

The external literature supplies important surrounding language. Finite prefixes belong to classical box-spline theory [3]; infinite Minkowski sums of matrix-dilated segments occur as infinite zonotopes in self-affine geometry [9]; and the finite zonotope volume formula is classical [5]. The contribution here is the Fabius–Rvachev synthesis: smooth convolution laws on those supports, exact matrix-Mahler and Rvachev equations, Thue–Morse derivative cubes, Bell–Bernoulli tensor resolvents, rotating geometric q -series, and the rational/irrational analytic dichotomies.

Status convention. **[repository input]** marks an imported fact already present in the repository or a classical ingredient used with citation. **[proved here]** marks a statement proved in this report and not located in the audited repository. **[derived synthesis]** marks a useful consequence or repackaging whose ingredients are standard. **[conjecture]** marks an unproved statement. Numerical evidence is always labeled **[numerical]** and is never used as a proof.

1.2 The scalar baseline

The probabilistic normalization behind Rvachev’s up-function is

$$X_2 = \sum_{n \geq 1} 2^{-n} U_n, \quad U_n \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1].$$

Its law has support $[-1, 1]$, density up, and characteristic function $\prod_{n \geq 1} \text{sinc}_{\text{rad}}(2^{-n}\xi)$, where throughout

$$\text{sinc}_{\text{rad}} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The bounded Fabius distribution on $[0, 1]$ is obtained by the affine change of variables $(X_2 + 1)/2$. The repository already derives its dyadic arithmetic, Thue–Morse formulas, exact moments, inverse asymptotics, and many transform representations. Our construction reduces to this law when $d = r = 1$, $A = 2$, and $v_1 = 1$.

1.3 Matrix notation

Fix integers $d, r \geq 1$. Let $A \in \text{GL}_d(\mathbb{R})$ be *expansive*: every eigenvalue of A has modulus greater than one. Put $B = A^{-1}$, let $V = [v_1 \cdots v_r] \in \mathbb{R}^{d \times r}$, and let $U_n = (U_{n,1}, \dots, U_{n,r})^T$ be independent vectors whose coordinates are independent $\text{Unif}[-1, 1]$ variables. Define

$$X = X_{A,V} := \sum_{n=1}^{\infty} B^n V U_n, \quad g_{n,j} := B^n v_j. \quad (1)$$

The law, characteristic function, moment generating function, and support will be denoted by μ , Φ , M , and K respectively. For $g \in \mathbb{R}^d$, $D_g = \sum_a g_a \partial_a$ is the directional derivative. Tensor powers are symmetric whenever they occur as moments or cumulants; Sym denotes normalized symmetrization.

2 The matrix-dilated law and its self-affine support

Theorem 2.1 (Existence, exact support, and affine recursion). *[proved here] The series (1) converges absolutely for every realization of the bounded digits. Its support is the compact centrally symmetric infinite zonotope*

$$K = \sum_{n \geq 1} \sum_{j=1}^r [-g_{n,j}, g_{n,j}], \quad (2)$$

and its support function is

$$h_K(w) = \sup_{x \in K} \langle w, x \rangle = \sum_{n \geq 1} \sum_{j=1}^r |\langle w, g_{n,j} \rangle|. \quad (3)$$

If X' is an independent copy of X and U is an independent uniform vector on $[-1, 1]^r$, then

$$X \stackrel{d}{=} B(VU + X'), \quad K = B(V[-1, 1]^r + K). \quad (4)$$

Finally, K has nonempty interior in $\mathbb{R}^d$ exactly when

$$\text{span}\{B^n v_j : n \geq 1, 1 \leq j \leq r\} = \mathbb{R}^d. \quad (5)$$

Proof. Choose ρ with $\rho(B) < \rho < 1$. There is C_ρ such that $\|B^n\| \leq C_\rho \rho^n$. Since $|U_{n,j}| \leq 1$,

$$\sum_{n,j} \|B^n v_j U_{n,j}\| \leq C_\rho \sum_j \|v_j\| \sum_{n \geq 1} \rho^n < \infty.$$

The convergence is therefore deterministic and uniform on the compact product space $\Omega = [-1, 1]^{\mathbb{N}_0 \times r}$.

The summation map $S : \Omega \rightarrow \mathbb{R}^d$ is continuous, and the product measure of the digits has full support Ω . Hence the support of its pushforward is the compact image $S(\Omega)$, which is exactly the Minkowski sum

(2). Support functions add under Minkowski addition and the support function of $[-g, g]$ is $|\langle w, g \rangle|$, proving (3). Splitting off the first scale gives

$$X = B \left(VU_1 + \sum_{m \geq 1} B^m VU_{m+1} \right),$$

which proves the distributional and set recursions.

If the displayed span is proper, K lies in that proper subspace. Conversely, if the span is all of $\mathbb{R}^d$, some finite subcollection of generators is a basis; their Minkowski sum contains a full-dimensional parallelootope, and therefore K has nonempty interior. $\square$

Remark 2.2. The support theorem is stronger than the usual convex-hull statement for a discrete self-affine attractor. Uniform digits already fill each segment, so the support itself, not merely its convex hull, is the infinite zonotope. In the one-generator case this support agrees with the infinite-zonotope convex hull appearing in self-affine two-digit geometry [9].

Corollary 2.3 (Exact projection support). *For every $w \in \mathbb{R}^d$, the scalar projection $\langle w, X \rangle$ has support $[-h_K(w), h_K(w)]$. In particular, widths and directional endpoint problems are explicit before the density itself is known.*

Proof. Project (2) under $x \mapsto \langle w, x \rangle$. The image of a generator segment $[-g, g]$ is

$$[-|\langle w, g \rangle|, |\langle w, g \rangle|].$$

Minkowski sums of centered intervals add their half-widths. $\square$

3 Fourier product, matrix Mahler equation, and smoothness

Theorem 3.1 (Entire sinc product and matrix Mahler equation). *[proved here] The characteristic function extends to an entire function on $\mathbb{C}^d$ and has the locally uniformly convergent product*

$$\Phi(z) = \mathbb{E}e^{i\langle z, X \rangle} = \prod_{n=1}^{\infty} \prod_{j=1}^r \text{sinc}_{\text{rad}}(\langle z, B^n v_j \rangle). \quad (6)$$

It satisfies

$$\Phi(A^T z) = m_V(z) \Phi(z), \quad m_V(z) := \prod_{j=1}^r \text{sinc}_{\text{rad}}(\langle z, v_j \rangle). \quad (7)$$

Moreover,

$$|\Phi(z)| \leq \exp h_K(-\text{Im } z), \quad (8)$$

so its exponential indicator is controlled exactly by the support zonoid.

Proof. Independence gives the finite products. Since $\text{sinc}_{\text{rad}} u = 1 - u^2/6 + O(u^4)$ locally and $\sum_{n,j} \|B^n v_j\|^2 < \infty$, the Weierstrass criterion gives local uniform convergence on $\mathbb{C}^d$ and hence an entire limit. Substitution gives

$$\Phi(A^T z) = \prod_{n,j} \text{sinc}_{\text{rad}}(\langle z, AB^n v_j \rangle) = \prod_j \text{sinc}_{\text{rad}}(\langle z, v_j \rangle) \prod_{n,j} \text{sinc}_{\text{rad}}(\langle z, B^n v_j \rangle).$$

Finally, $|e^{i\langle z, x \rangle}| = e^{-\langle \text{Im } z, x \rangle}$ and $x \in K$ almost surely, so expectation is bounded by the maximum over K , which is (8). $\square$

3.1 A finite-frame decay mechanism

The one-dimensional up-product decays roughly like a Gaussian in $\log |\xi|$. The same mechanism persists without a scalar order on the generators. A finite spanning block acts as a frame, and every matrix-dilated block contributes at least one large sinc argument.

Lemma 3.2 (Block-frame estimate). *Assume (5). Then there are an integer $L \geq 1$ and $c > 0$ such that*

$$\max_{1 \leq k \leq L, 1 \leq j \leq r} \left| \langle y, B^k v_j \rangle \right| \geq c \|y\| \quad (y \in \mathbb{R}^d). \quad (9)$$

For every $\beta > \rho(A)$ there is $C_\beta \geq 1$ such that, for every block index $m \geq 0$ and every $\xi \in \mathbb{R}^d$, one of the factors with $n = mL + k$, $1 \leq k \leq L$, has argument at least

$$cC_\beta^{-1} \beta^{-mL} \|\xi\|. \quad (10)$$

Proof. Choose L so that the finite family in (9) spans $\mathbb{R}^d$. The maximum on the left, restricted to the unit sphere, is continuous and strictly positive, giving c . Since $\beta > \rho(A)$, $\|A^m\| \leq C_\beta \beta^m$. Apply (9) to $y = B^{TmL} \xi$. Its norm is bounded below by $\|\xi\| / \|A^{TmL}\| \geq C_\beta^{-1} \beta^{-mL} \|\xi\|$, and

$$\langle B^{TmL} \xi, B^k v_j \rangle = \langle \xi, B^{mL+k} v_j \rangle.$$

□

Theorem 3.3 (Sharp dimensional dichotomy and log-Gaussian decay). *[proved here] Let $W = \text{span}\{B^n v_j : n \geq 1, 1 \leq j \leq r\}$. If $W \neq \mathbb{R}^d$, then μ is supported on a proper subspace and is singular with respect to d -dimensional Lebesgue measure. If $W = \mathbb{R}^d$, then μ has a compactly supported C^∞ density f , and for every $\beta > \rho(A)$,*

$$|\Phi(\xi)| \leq \exp \left\{ -\frac{(\log \|\xi\|)^2}{2L \log \beta} + O(\log \|\xi\|) \right\} \quad (\|\xi\| \rightarrow \infty), \quad (11)$$

where L is any spanning-block length in Lemma 3.2. In particular, $\|\xi\|^N \Phi(\xi)$ is bounded for every N .

Proof. Only the full-span case remains. On the real axis, $|\text{sinc}_{\text{rad}} t| \leq \min(1, |t|^{-1})$. Choose from each block the factor supplied by Lemma 3.2. Put $R = \|\xi\|$ and let M be the largest integer for which $cC_\beta^{-1} \beta^{-ML} R \geq 1$. Discarding all factors except one from each block $0 \leq m \leq M$ gives

$$|\Phi(\xi)| \leq \prod_{m=0}^M \frac{C_\beta \beta^{mL}}{cR}.$$

Taking logarithms,

$$\log |\Phi(\xi)| \leq -(M+1) \log \frac{cR}{C_\beta} + \frac{L \log \beta}{2} M(M+1).$$

Since $M = (\log R)/(L \log \beta) + O(1)$, this is (11). The bound is faster than every negative power, so $\xi \mapsto \xi^\alpha \Phi(\xi)$ is integrable for every multi-index α . Fourier inversion produces a C^∞ density. Its support is K by Theorem 2.1. □

Remark 3.4 (Non-normal matrices). No diagonalization, normality, or real spectrum is used. The constant in (11) records two genuinely matrix effects: the number L of scales needed to span, and a growth envelope β for A^m . Sharpening the bound for non-normal A should involve singular-value Lyapunov exponents rather than the spectral-radius envelope; see Open problem 14.1.

3.2 Log-concavity survives the infinite matrix cascade

Theorem 3.5 (Log-concavity and convex superlevel sets). *[proved here] The probability measure μ is log-concave in the measure-theoretic sense. Under the span condition, its smooth density f is log-concave on $\mathbb{R}^d$, is strictly positive on $\text{int } K$, and vanishes off K . Consequently every positive superlevel set $\{x : f(x) \geq c\}$ is compact and convex.*

Proof. Uniform probability on a segment is log-concave as a measure. Affine images, finite convolutions, and weak limits preserve log-concavity; these are standard consequences of the Prékopa–Leindler theory [8]. Hence every finite-prefix law μ_N is log-concave and, because $\mu_N \Rightarrow \mu$, so is μ .

When the generators span, Theorem 3.3 gives a full-dimensional continuous density. The density characterization of a full-dimensional log-concave measure supplies a log-concave representative. It agrees almost everywhere with f , hence everywhere on $\text{int } K$ by continuity; outside K both vanish, and continuity forces the boundary value to be zero. The positivity set of a full-dimensional log-concave density is the interior of its convex support. Convexity of positive superlevel sets is the defining pointwise log-concavity inequality. $\square$

Remark 3.6. Smoothness does not imply strict log-concavity. A sufficiently dominant finite prefix can leave a genuine flat core after convolution with the remaining tail. The rotating family below makes this mechanism quantitative in Proposition 10.3.

4 The matrix Rvachev equation

Let σ_g be the uniform probability measure on the segment $[-g, g]$. As a distribution,

$$D_g \sigma_g = \frac{1}{2}(\delta_{-g} - \delta_g). \quad (12)$$

This elementary endpoint identity is the source of both Rvachev’s differential law and the Thue–Morse cube below.

Theorem 4.1 (One-scale matrix Rvachev equation). *[proved here] Assume the span condition, and let f be the density from Theorem 3.3. Then*

$$\left(\prod_{j=1}^r D_{Bv_j} \right) f(x) = \frac{|\det A|}{2^r} \sum_{\omega \in \{0,1\}^r} (-1)^{|\omega|} f(Ax - V(2\omega - \mathbf{1})). \quad (13)$$

For $d = r = 1$, $A = 2$, and $V = (1)$, it becomes

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)). \quad (14)$$

Proof. The one-step recursion factors the law as

$$\mu = \left(\ast_{j=1}^r \sigma_{Bv_j} \right) \ast (B_{\#} \mu).$$

Apply one directional derivative to its matching segment factor and use (12). Expanding the resulting r endpoint choices gives translations by $BV(2\omega - \mathbf{1})$ with signs $(-1)^{|\omega|}$. The density of $B_{\#} \mu$ is $|\det A|f(Ax)$, so translating and simplifying produces (13). In the scalar case the left side is $D_{1/2} \text{up} = \text{up}'/2$, and the displayed specialization follows. $\square$

Remark 4.2. Equation (13) is a functional–differential equation on an anisotropic expanding geometry. The finite difference on the right is the r -dimensional cube difference generated by the columns of V ; the left is the matching mixed directional derivative at the contracted scale. It is therefore a direct matrix analogue of the defining Rvachev mechanism, not merely a tensor product of one-dimensional equations.

5 Finite box splines and all-orders geometric approximation

Define

$$X_N = \sum_{n=1}^N B^n V U_n, \quad T_N = X - X_N, \quad K_N = \sum_{n \leq N, j} [-g_{n,j}, g_{n,j}]. \quad (15)$$

The law of X_N is the centered box spline associated with the multiset of rN generators $g_{n,j}$; its Fourier transform is the corresponding finite sinc product. This is the standard probabilistic realization of a box spline [3].

Proposition 5.1 (Exact head–tail self-similarity). *[proved here] One has*

$$T_N \stackrel{d}{=} B^N X', \quad \mu = \mu_N * (B_{\#}^N \mu), \quad K = K_N + B^N K, \quad (16)$$

where X' is an independent copy of X . Consequently, for every $p \geq 1$,

$$W_p(\mu, \mu_N) \leq \left(\mathbb{E} \|B^N X\|^p \right)^{1/p} \leq \sum_{n > N, j} \|g_{n,j}\|, \quad (17)$$

and the same last quantity bounds the Hausdorff distance $d_H(K, K_N)$.

Proof. Reindex the tail and use the same digit law. The coupling (X, X_N) gives the Wasserstein estimate, while the deterministic tail norm gives the final bound. The support identity follows from Minkowski addition. $\square$

For the differential expansion define the tail power tensors

$$Q_{k,N} := \sum_{n > N} \sum_{j=1}^r g_{n,j}^{\otimes k}, \quad S_{k,N} := \sum_{n > N, j} \|g_{n,j}\|^k. \quad (18)$$

Thus $Q_{2,N}$ is three times the tail covariance. If $\rho(B) < \rho < 1$, then $S_{k,N} = O(\rho^{kN})$.

Theorem 5.2 (C^m convergence and the first two correction layers). *[proved here] Assume the span condition. For every fixed $m \geq 0$, the finite box-spline density f_N is C^{m+6} once N is sufficiently large, and*

$$f = f_N + \frac{1}{6} Q_{2,N} : D^2 f_N + O_{C^m}(S_{2,N}^2), \quad (19)$$

$$f = f_N + \frac{1}{6} Q_{2,N} : D^2 f_N - \frac{1}{180} \sum_{n > N, j} D_{g_{n,j}}^4 f_N + \frac{1}{72} (Q_{2,N} : D^2)^2 f_N + O_{C^m}(S_{2,N}^3). \quad (20)$$

Here $Q : D^2 = \sum_{a,b} Q_{ab} \partial_a \partial_b$. In particular, $\|f - f_N\|_{C^m} = O(\rho^{2N})$ for every $\rho > \rho(B)$.

Proof. Choose a fixed prefix N_0 containing sufficiently many spanning blocks that $|\Phi_{N_0}(\xi)| \leq C(1 + \|\xi\|)^{-m-2k-d-2}$ for the needed value of k . The block argument of Theorem 3.3, stopped after finitely many blocks, gives such a prefix. For $N \geq N_0$, every additional real sinc factor has modulus at most one, so Φ_N inherits this integrable majorant.

The tail multiplier is

$$\Psi_N(\xi) = \prod_{n > N, j} \text{sinc}_{\text{rad}} \langle \xi, g_{n,j} \rangle.$$

The cumulant expansion of a uniform digit begins

$$\log \text{sinc}_{\text{rad}} t = -\frac{t^2}{6} - \frac{t^4}{180} + O(t^6).$$

Therefore

$$\Psi_N(\xi) = 1 - \frac{1}{6}Q_{2,N}[\xi, \xi] - \frac{1}{180} \sum_{n>N,j} \langle \xi, g_{n,j} \rangle^4 + \frac{1}{72}Q_{2,N}[\xi, \xi]^2 + O(\|\xi\|^6 S_{2,N}^3).$$

The same estimate can be made global by increasing the constant, because $|1 - \text{sinc}_{\text{rad}} t| \leq Ct^2$ and the sixth-order Taylor remainder is bounded by $C|t|^6$. Multiply by Φ_N , integrate against $\|\xi\|^m$, and use the fixed integrable majorant. Under the convention $\widehat{f}_{\text{ang}}(-\xi) = \int e^{i\langle \xi, x \rangle} f(x) dx$, multiplication by $-\langle \xi, g \rangle^2$ is the Fourier image of D_g^2 , giving (19) and (20). $\square$

5.1 The Bell–Bernoulli operator to arbitrary order

Let

$$c_{2m} := \kappa_{2m}(U) = \frac{2^{2m-1} B_{2m}}{m}, \quad c_{2m+1} = 0, \quad (21)$$

for $U \sim \text{Unif}[-1, 1]$. The tail multiplier has the exact local logarithm

$$\log \Psi_N(\xi) = \sum_{m \geq 1} \frac{c_{2m}}{(2m)!} (i)^{2m} \sum_{n>N,j} \langle \xi, g_{n,j} \rangle^{2m}. \quad (22)$$

Thus the formal differential operator

$$\mathcal{L}_N = \sum_{m \geq 1} \frac{c_{2m}}{(2m)!} \sum_{n>N,j} D_{g_{n,j}}^{2m} \quad (23)$$

obeys $f \sim \exp(\mathcal{L}_N)f_N$. Expanding the exponential uses complete Bell polynomials. The proof of Theorem 5.2 extends verbatim: after retaining all terms of geometric degree at most $2M$, the C^m remainder is $O(\rho^{(2M+2)N})$. This is a matrix-valued counterpart of the Bell–Bernoulli finite-sinc expansions already developed in the scalar repository.

6 The multivariate Thue–Morse derivative cube

Order the first N scales' generators as $g_1, \dots, g_M$, where $M = rN$. For $\omega = (\omega_1, \dots, \omega_M) \in \{0, 1\}^M$, put

$$s_\omega = \sum_{\ell=1}^M (2\omega_\ell - 1)g_\ell, \quad \varepsilon_\omega = (-1)^{|\omega|}. \quad (24)$$

When the bit strings are ordered by their binary integers, ε_ω is the Prouhet–Thue–Morse sign.

Theorem 6.1 (Exact derivative cube). *[proved here] In the sense of distributions,*

$$\left(\prod_{\ell=1}^M D_{g_\ell} \right) \mu = 2^{-M} \sum_{\omega \in \{0,1\}^M} \varepsilon_\omega \tau_{s_\omega} (B_{\#}^N \mu), \quad (25)$$

where $\tau_s \nu$ denotes translation by s . Under the span condition this is the pointwise smooth identity

$$\left(\prod_{\ell=1}^M D_{g_\ell} \right) f(x) = \frac{|\det A|^N}{2^M} \sum_{\omega \in \{0,1\}^M} \varepsilon_\omega f(A^N(x - s_\omega)). \quad (26)$$

Proof. Factor μ into the M segment measures and the tail $B_{\#}^N \mu$. Assign D_{g_ℓ} to the matching segment measure and use (12). Expanding the endpoint choices gives (25). The tail density is $|\det A|^N f(A^N x)$, yielding (26). $\square$

Corollary 6.2 (Tensor Prouhet cancellation). *[proved here] The cube points satisfy the exponential identity*

$$\sum_{\omega} \varepsilon_{\omega} e^{\langle z, s_{\omega} \rangle} = (-2)^M \prod_{\ell=1}^M \sinh \langle z, g_{\ell} \rangle. \quad (27)$$

Consequently,

$$\sum_{\omega} \varepsilon_{\omega} s_{\omega}^{\otimes k} = 0 \quad (0 \leq k < M), \quad (28)$$

and at the first surviving degree

$$\sum_{\omega} \varepsilon_{\omega} s_{\omega}^{\otimes M} = (-2)^M M! \operatorname{Sym}(g_1 \otimes \cdots \otimes g_M). \quad (29)$$

Equivalently, the signed cube annihilates every polynomial on $\mathbb{R}^d$ of total degree less than M .

Proof. The left side factors coordinate by coordinate as

$$\prod_{\ell=1}^M \left(e^{-\langle z, g_{\ell} \rangle} - e^{\langle z, g_{\ell} \rangle} \right),$$

which is (27). Compare homogeneous Taylor terms at $z = 0$. The product of M hyperbolic sines has no term below degree M , and its degree- M term is $\prod_{\ell} \langle z, g_{\ell} \rangle$. Polarization gives (29). $\square$

Remark 6.3 (Why this is genuinely multivariate). The signs are still the one-dimensional Thue–Morse character of the Boolean cube, but the nodes are not a scalar dyadic grid. They are vertices of an anisotropic matrix-generated zonotope, and the first nonzero tensor records the entire ordered generator geometry. Tensor-product Fabius laws appear only when A and V split into invariant coordinate blocks.

7 Bell–Bernoulli tensor cumulants and arithmetic

For t near zero, the cumulant generating function is

$$\log M(t) = \sum_{n,j} \log \frac{\sinh \langle t, g_{n,j} \rangle}{\langle t, g_{n,j} \rangle}. \quad (30)$$

The scalar Bernoulli expansion in (21) can be summed exactly over matrix scales.

Theorem 7.1 (Tensor resolvent formula). *[proved here] Every odd cumulant tensor of X vanishes. For $m \geq 1$,*

$$\kappa_{2m}(X) = c_{2m} (A^{\otimes 2m} - I)^{-1} \sum_{j=1}^r v_j^{\otimes 2m}. \quad (31)$$

Equivalently, it is the unique symmetric tensor solving the higher Lyapunov equation

$$(A^{\otimes 2m} - I) \kappa_{2m}(X) = c_{2m} \sum_j v_j^{\otimes 2m}. \quad (32)$$

In particular, the covariance matrix Σ is the unique solution of

$$A \Sigma A^T - \Sigma = \frac{1}{3} V V^T. \quad (33)$$

Proof. Cumulants add over independent summands and transform tensorially under linear maps. Hence

$$\kappa_k(X) = c_k \sum_{n \geq 1} \sum_j (B^n v_j)^{\otimes k} = c_k \sum_{n \geq 1} (B^{\otimes k})^n \sum_j v_j^{\otimes k}.$$

All eigenvalues of $B^{\otimes k}$ have modulus below one, so the geometric operator series equals

$$(I - B^{\otimes k})^{-1} B^{\otimes k} = (A^{\otimes k} - I)^{-1}.$$

This proves (31). For $k = 2$, interpret the tensor equation as a matrix identity and use $c_2 = 1/3$. $\square$

Corollary 7.2 (All joint moments by set partitions). *[proved here] Let Π_k be the set of set partitions of $\{1, \dots, k\}$. Then*

$$\mathbb{E}X^{\otimes k} = \sum_{\pi \in \Pi_k} \text{Sym} \bigotimes_{C \in \pi} \kappa_{|C|}(X). \quad (34)$$

For a projection $Y_w = \langle w, X \rangle$, put $\lambda_k(w) = \langle \kappa_k(X), w^{\otimes k} \rangle$. Its scalar moments are the complete Bell polynomials

$$\mathbb{E}Y_w^k = \mathcal{B}_k(\lambda_1(w), \dots, \lambda_k(w)). \quad (35)$$

Proof. This is the classical moment–cumulant relation, written in symmetric tensor notation; see McCullagh’s tensor treatment [7]. Contracting with $w^{\otimes k}$ gives the scalar complete Bell polynomial. $\square$

Corollary 7.3 (Rational and algebraic arithmetic). *[proved here] If A and V have entries in a number field K_0 , then every joint moment of X lies in K_0 . In particular, rational data give rational moments. If A and V are integral, every coordinate of*

$$\frac{\det(A^{\otimes 2m} - I)}{c_{2m}} \kappa_{2m}(X) \quad (36)$$

is an integer. Thus the denominator of each order-2m cumulant divides the Bernoulli denominator of c_{2m} times $|\det(A^{\otimes 2m} - I)|$; Bell partitions give an explicit, though generally non-sharp, denominator bound for every moment.

Proof. The matrix in (31) is invertible because every eigenvalue of $A^{\otimes k}$ has modulus greater than one. Field closure under inversion proves the first assertion. For integral data, use the adjugate formula for the inverse of $A^{\otimes 2m} - I$ and then (34). $\square$

Remark 7.4. The scalar dyadic moment denominators in the repository possess far subtler 2-adic structure than this generic determinant bound. The matrix formula identifies the new arithmetic object that should replace the factors $2^{2m} - 1$: the determinant and invariant factors of $A^{\otimes 2m} - I$. Smith normal form, rather than a single valuation, is the natural next layer.

8 Appell–Koopman structure and Legendre projections

8.1 A matrix Appell intertwiner

Define the averaging operator on polynomials by

$$(Tp)(x) = \mathbb{E} p(B(x + VU)), \quad (37)$$

where U is uniform on $[-1, 1]^r$. Define the multivariate Appell generating function

$$\mathcal{A}(x, t) = \frac{e^{\langle t, x \rangle}}{M(t)} = \sum_{k \geq 0} \frac{1}{k!} \langle \mathbf{A}_k(x), t^{\otimes k} \rangle. \quad (38)$$

The tensors $\mathbf{A}_k$ are polynomial-valued and begin with $\mathbf{A}_0 = 1$, $\mathbf{A}_1 = x$, and $\mathbf{A}_2 = x^{\otimes 2} - \Sigma$.

Theorem 8.1 (Exact Appell–Koopman intertwining). *[proved here] The generating function obeys*

$$T\mathcal{A}(x, t) = \mathcal{A}(x, B^T t), \quad (39)$$

and therefore

$$T\mathbf{A}_k = (B^{\otimes k})\mathbf{A}_k. \quad (40)$$

If A is diagonalizable over $\mathbb{C}$, contraction of $\mathbf{A}_k$ with symmetric tensor products of left eigenvectors gives a complete Appell eigenbasis for polynomials. The degree- k eigenvalues are products of k eigenvalues of $B = A^{-1}$.

Proof. The affine fixed-point equation implies

$$M(t) = M(B^T t) \prod_{j=1}^r \frac{\sinh \langle B^T t, v_j \rangle}{\langle B^T t, v_j \rangle}. \quad (41)$$

Using this identity in (37),

$$T\mathcal{A}(x, t) = \frac{e^{\langle B^T t, x \rangle}}{M(t)} \prod_j \frac{\sinh \langle B^T t, v_j \rangle}{\langle B^T t, v_j \rangle} = \frac{e^{\langle B^T t, x \rangle}}{M(B^T t)}.$$

Compare homogeneous coefficients in t . Since Appell tensors are monic by (38), they form a triangular basis of the polynomial algebra; diagonalizing B gives the final assertion. For background on Appell families and their multivariate relatives, see [1]. $\square$

8.2 Exact Legendre coefficients of every projection

Fix $w \neq 0$, put $h = h_K(w)$, and normalize $Y = \langle w, X \rangle / h \in [-1, 1]$. Under the span condition, Y has a smooth compactly supported density p_w on $[-1, 1]$. Write

$$p_w(y) = \sum_{\ell=0}^{\infty} a_{\ell}(w) P_{\ell}(y) \quad (42)$$

in $L^2[-1, 1]$ and, in fact, with rapid coefficient decay.

Proposition 8.2 (Bell–Bernoulli–Legendre bridge). *[proved here] The exact coefficient is*

$$a_{\ell}(w) = \frac{2\ell + 1}{2} \mathbb{E} P_{\ell}(Y). \quad (43)$$

Using

$$P_{\ell}(y) = 2^{-\ell} \sum_{k=0}^{\lfloor \ell/2 \rfloor} (-1)^k \frac{(2\ell - 2k)!}{k!(\ell - k)!(\ell - 2k)!} y^{\ell - 2k}, \quad (44)$$

all coefficients are finite expressions in the tensor resolvents (31) and complete Bell polynomials (35). Every odd coefficient vanishes. The first even ones are

$$\begin{aligned} a_0 &= \frac{1}{2}, \\ a_2 &= \frac{5}{4} (3\mathbb{E} Y^2 - 1), \\ a_4 &= \frac{9}{16} (35\mathbb{E} Y^4 - 30\mathbb{E} Y^2 + 3). \end{aligned} \quad (45)$$

Proof. Orthogonality gives (43); insert (44) and use Corollary 7.2. The law is centrally symmetric, so odd moments and odd Legendre coefficients vanish. $\square$

Remark 8.3. This bridge does not assert that the multivariate density itself has a preferred basis of products of Legendre polynomials. It says something more invariant: every Radon projection has an exact scalar Legendre expansion whose coefficients are contractions of matrix resolvents. The direction w enters through both the support gauge $h_K(w)$ and the tensor contractions.

9 Zonoid volume and the geometry carried by the density

Theorem 9.1 (Infinite-zonotope volume formula). *[proved here] Enumerate all generators in (2) as $(g_\ell)_{\ell \geq 1}$. Then*

$$\text{vol}_d(K) = 2^d \sum_{1 \leq \ell_1 < \dots < \ell_d} |\det(g_{\ell_1}, \dots, g_{\ell_d})|. \quad (46)$$

The series converges absolutely. It is positive exactly under the span condition.

Proof. For the finite zonotope generated by $g_1, \dots, g_M$, the classical zonotope formula is

$$2^d \sum_{\ell_1 < \dots < \ell_d} |\det(g_{\ell_1}, \dots, g_{\ell_d})|;$$

see [5]. Absolute convergence follows from $|\det(g_{\ell_1}, \dots, g_{\ell_d})| \leq \prod_i \|g_{\ell_i}\|$ and $\sum_\ell \|g_\ell\| < \infty$. The finite zonotopes converge to K in Hausdorff distance, and volume is continuous under Hausdorff convergence of convex compact sets. Positivity is equivalent to the existence of a nonzero $d \times d$ minor, hence to spanning. $\square$

Corollary 9.2 (A determinant series for matrix-dilated support volume). *For (1),*

$$\text{vol}_d(K) = 2^d \sum_{(n_1, j_1) < \dots < (n_d, j_d)} |\det(B^{n_1} v_{j_1}, \dots, B^{n_d} v_{j_d})|. \quad (47)$$

Thus support volume is a convergent matrix q -series in the singular geometry of powers of B .

10 The rotating planar family

Set $d = 2$, $r = 1$, choose $0 < q < 1$, an angle θ , and a nonzero vector v . Let R_θ denote counterclockwise rotation and define

$$X_{q,\theta} = \sum_{n \geq 1} q^n R_{n\theta} v U_n. \quad (48)$$

This is the matrix law with $B = qR_\theta$ and $A = q^{-1}R_{-\theta}$. It is collinear only when $\theta \in \pi\mathbb{Z}$; otherwise it has a C^∞ density.

Proposition 10.1 (Rotating Mahler and Rvachev equations). *[proved here] The characteristic function and density satisfy*

$$\Phi(q^{-1}R_\theta \xi) = \text{sinc}_{\text{rad}} \langle \xi, v \rangle \Phi(\xi), \quad (49)$$

$$D_{qR_\theta v} f(x) = \frac{1}{2q^2} [f(q^{-1}R_{-\theta}x + v) - f(q^{-1}R_{-\theta}x - v)]. \quad (50)$$

Proof. Specialize Theorems 3.1 and 4.1, noting that $|\det A| = q^{-2}$. $\square$

10.1 Perimeter and area as exact q -series

Write $a = \|v\|$. The support is

$$K_{q,\theta} = \sum_{n \geq 1} [-q^n R_{n\theta} v, q^n R_{n\theta} v]. \quad (51)$$

Theorem 10.2 (Exact support function, perimeter, and area). *[proved here] If $w_\varphi = (\cos \varphi, \sin \varphi)$ and v has angle φ_0 , then*

$$h_{q,\theta}(\varphi) = a \sum_{n \geq 1} q^n |\cos(n\theta + \varphi_0 - \varphi)|. \quad (52)$$

The perimeter is independent of θ :

$$\text{Per}(K_{q,\theta}) = \frac{4aq}{1-q}. \quad (53)$$

The area is

$$\boxed{\text{Area}(K_{q,\theta}) = \frac{4a^2 q^2}{1-q^2} \sum_{m=1}^{\infty} |\sin(m\theta)| q^m.} \quad (54)$$

The N -stage formulas are

$$\text{Per}(K_N) = \frac{4aq(1-q^N)}{1-q}, \quad (55)$$

$$\text{Area}(K_N) = \frac{4a^2 q^2}{1-q^2} \sum_{m=1}^{N-1} q^m |\sin(m\theta)| (1-q^{2(N-m)}). \quad (56)$$

Proof. The support function is (3). In the plane, the perimeter of a Minkowski sum is the sum of perimeters, and the degenerate segment $[-g, g]$ contributes $4\|g\|$; approximation by thin rectangles justifies this convention directly. Summing $4aq^n$ gives (53).

The two-dimensional case of (46) gives

$$\text{Area}(K) = 4 \sum_{1 \leq n < k} q^{n+k} a^2 |\sin((k-n)\theta)|.$$

Group by $m = k - n$ and sum $\sum_{n \geq 1} q^{2n} = q^2/(1-q^2)$ to obtain (54). Stopping at $k \leq N$ gives $\sum_{n=1}^{N-m} q^{2n} = q^2(1-q^{2(N-m)})/(1-q^2)$ and hence (56). $\square$

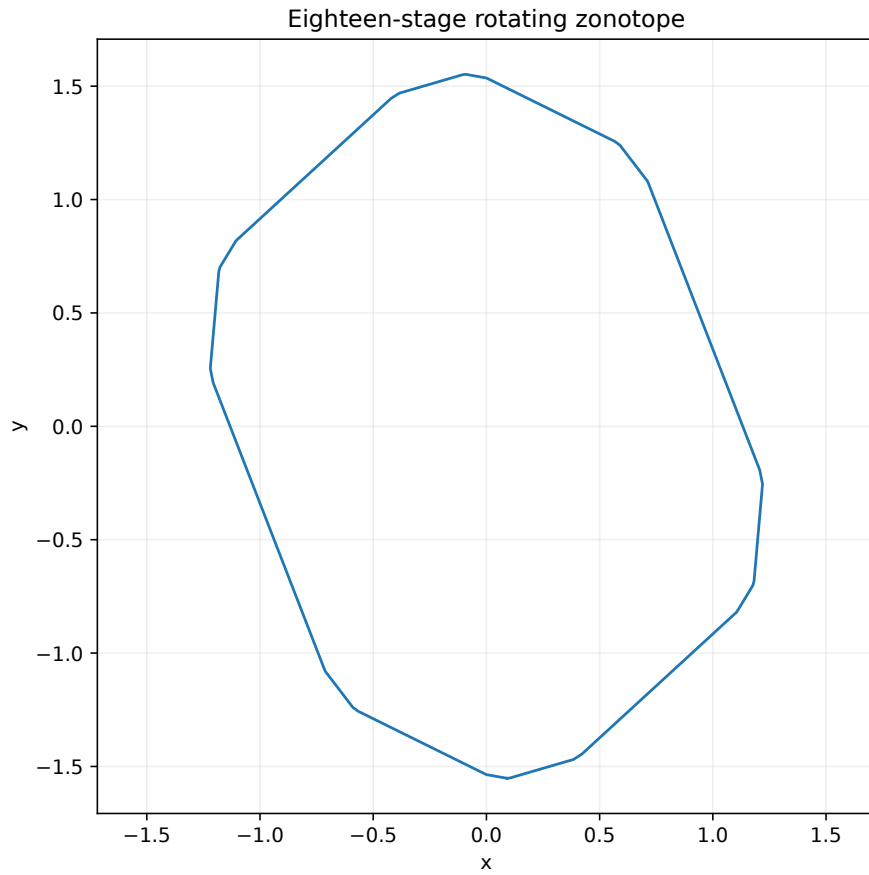


Figure 1: The 18-stage support for $q = 0.68$ and $\theta/\pi = (\sqrt{5} - 1)/2$. The finite polygon already approximates an infinite zonotope with dense edge directions.

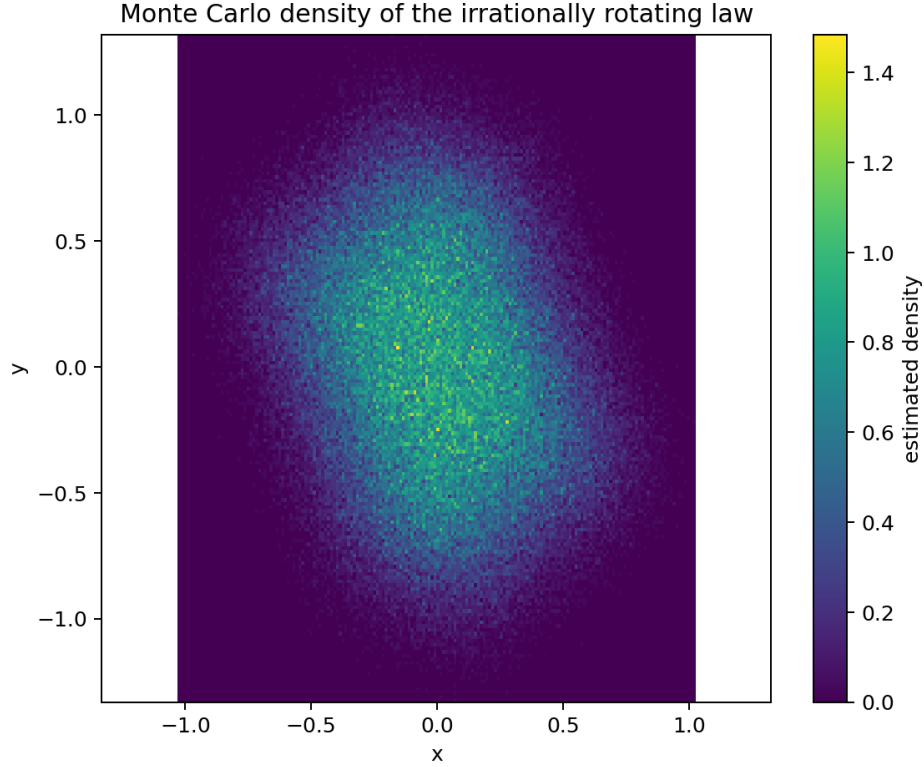


Figure 2: Monte Carlo density for the same irrationally rotating law. The support boundary is infinitely polygonal, while the density is C^∞ by Theorem 3.3 and log-concave by Theorem 3.5.

Proposition 10.3 (A certified central-plateau regime). *[proved here] Assume $\sin \theta \neq 0$. Put*

$$P_2 = [-qR_\theta v, qR_\theta v] + [-q^2 R_{2\theta} v, q^2 R_{2\theta} v], \quad K_{>2} = \sum_{n \geq 3} [-q^n R_{n\theta} v, q^n R_{n\theta} v],$$

and define the Minkowski erosion $P_2 \ominus K_{>2} = \{x : x - K_{>2} \subseteq P_2\}$. On this erosion the density is exactly constant:

$$f(x) = \frac{1}{4a^2 q^3 |\sin \theta|}. \quad (57)$$

In particular, if

$$\frac{q}{1-q} < |\sin \theta|, \quad (58)$$

then the erosion contains a neighborhood of the origin, so f has a full-dimensional central plateau and cannot be strictly log-concave.

Proof. The first two digits have the uniform density $(4|\det(qR_\theta v, q^2 R_{2\theta} v)|)^{-1}$ on the parallelogram P_2 . Convolution with the tail measure therefore leaves that constant unchanged at all x for which $x - K_{>2} \subseteq P_2$, proving (57). The Euclidean inradius of P_2 is $aq^2 |\sin \theta|$, while $K_{>2}$ lies in the ball of radius $aq^3/(1-q)$. The strict inequality (58) places the latter ball strictly inside P_2 and makes the erosion have nonempty interior. $\square$

10.2 Dense edges and an exact curvature measure

Theorem 10.4 (Dense-edge boundary of the irrational support). *[proved here] Suppose θ/π is irrational. For every $n \geq 1$, the boundary of $K_{q,\theta}$ has exactly two opposite exposed edges parallel to $R_{n\theta} v$, each of length $2aq^n$. There are no other nontrivial exposed faces. Their directions are dense modulo π , and the union of their relative interiors has full arclength measure on $\partial K_{q,\theta}$.*

Proof. For a unit normal u , the exposed face of a Minkowski sum is the Minkowski sum of the exposed faces of its summands. A segment $[-g, g]$ contributes an endpoint unless $u \perp g$, in which case it contributes the full segment. Irrationality ensures that no two generator directions agree modulo π . Thus a nontrivial face occurs exactly when u is perpendicular to one generator, and its length is $2 \|g_n\| = 2aq^n$. The directions $n\theta \bmod \pi$ are dense by Weyl's equidistribution theorem [10]. The two edges for each n have total length $4aq^n$; summing gives $4aq/(1 - q)$, exactly the perimeter in (53). Hence the complement has zero arclength. $\square$

Let S_K be the planar surface-area measure: an edge contributes its length at its outward unit normal. The proof gives the atomic measure

$$S_{K_{q,\theta}} = 2a \sum_{n \geq 1} q^n (\delta_{n\theta + \varphi_0 + \pi/2} + \delta_{n\theta + \varphi_0 - \pi/2}). \quad (59)$$

With the unnormalized angular Fourier–Stieltjes convention $\text{FSU}_{2\pi}[S][k] = \int e^{-ik\phi} dS(\phi)$,

$$\text{FSU}_{2\pi}[S][k] = 4a \cos \frac{k\pi}{2} e^{-ik\varphi_0} \frac{qe^{-ik\theta}}{1 - qe^{-ik\theta}}. \quad (60)$$

Thus the boundary's curvature data are finite cyclotomic resolvents even though the density inside is governed by an infinite sinc product.

10.3 Rational functions versus natural boundaries

The area series introduces

$$S_\theta(z) = \sum_{m \geq 1} |\sin(m\theta)| z^m, \quad |z| < 1. \quad (61)$$

Theorem 10.5 (Rotation-orbit Abel residues). *[proved here] Let $g(x) = \sum_{k \in \mathbb{Z}} c_k e^{ikx}$ be a continuous periodic function with $\sum_k |c_k| < \infty$, let $\theta/(2\pi)$ be irrational, and put*

$$G(z) = \sum_{n \geq 1} g(n\theta + \delta) z^n. \quad (62)$$

Then for every $k \in \mathbb{Z}$,

$$\lim_{r \uparrow 1} (1 - r)G(re^{-ik\theta}) = c_k e^{ik\delta}. \quad (63)$$

Every point $e^{-ik\theta}$ for which $c_k \neq 0$ is therefore a singularity of G . If those points are dense on the unit circle, the circle is a natural boundary.

Proof. Insert the absolutely convergent Fourier series:

$$(1 - r)G(re^{-ik\theta}) = \sum_{\ell \in \mathbb{Z}} c_\ell e^{i\ell\delta} \frac{(1 - r)re^{i(\ell-k)\theta}}{1 - re^{i(\ell-k)\theta}}.$$

The $\ell = k$ term tends to $c_k e^{ik\delta}$, and every other term tends to zero because irrationality prevents a unit denominator. Each fraction has modulus at most one, so dominated convergence applies using $\sum |c_\ell| < \infty$. If G extended holomorphically across the candidate point, it would be bounded there and the left side would tend to zero, a contradiction. Dense singularities exclude continuation across every boundary arc. $\square$

The absolutely convergent Fourier series

$$|\sin x| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{k \geq 1} \frac{\cos(2kx)}{4k^2 - 1} \quad (64)$$

contains a nonzero coefficient at every even frequency.

Corollary 10.6 (Arithmetic dichotomy for rotating area). *[proved here] If $\theta/\pi \in \mathbb{Q}$, the coefficient sequence $|\sin(m\theta)|$ is periodic, so $S_\theta(z)$ is rational; if $\theta = \pi a/p$ with integers a, p , one may write*

$$S_\theta(z) = \frac{\sum_{m=1}^p |\sin(m\theta)| z^m}{1 - z^p}. \quad (65)$$

If $\theta/\pi \notin \mathbb{Q}$, the unit circle is a natural boundary of $S_\theta(z)$ and hence of the analytically continued area in (54).

Proof. The rational case is periodic summation. In the irrational case apply Theorem 10.5 to (64). The singular points $e^{-2ik\theta}$ are dense because θ/π is irrational. The rational prefactor in (54) cannot remove a dense set of nonzero Abel residues. $\square$

Corollary 10.7 (Natural boundary of every directional support series). *[proved here] For irrational θ/π and every fixed direction φ , the analytic function*

$$H_{\theta,\varphi}(z) = \sum_{n \geq 1} |\cos(n\theta + \varphi_0 - \varphi)| z^n \quad (66)$$

has the unit circle as a natural boundary.

Proof. The Fourier series of $|\cos x|$ is a phase shift of (64) and has nonzero coefficients at all even frequencies. Apply Theorem 10.5. $\square$

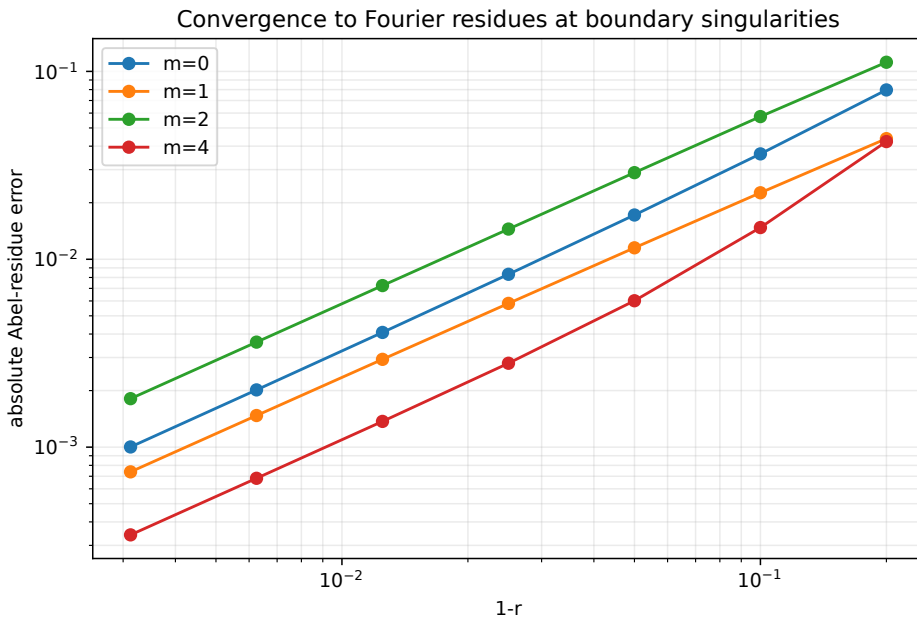


Figure 3: Numerical convergence of the Abel residues in (63) for the area series at four dense boundary singularities. The limiting constants are the exact Fourier coefficients in (64).

10.4 An Abel shape theorem: polygons versus a disk

The natural normalization is

$$\tilde{K}_{q,\theta} := \frac{1-q}{q} K_{q,\theta}; \quad (67)$$

its perimeter is the constant $4a$.

Theorem 10.8 (Abel shape limit). *[proved here]* As $q \uparrow 1$:

1. if θ/π is irrational, then

$$\tilde{K}_{q,\theta} \longrightarrow \{x \in \mathbb{R}^2 : \|x\| \leq 2a/\pi\} \quad (68)$$

in Hausdorff distance;

2. if θ/π is rational and p is a period modulo π , then

$$\tilde{K}_{q,\theta} \longrightarrow \frac{1}{p} \sum_{m=1}^p [-R_{m\theta}v, R_{m\theta}v], \quad (69)$$

a finite zonotope.

In the irrational case,

$$\left(\frac{1-q}{q}\right)^2 \text{Area}(K_{q,\theta}) \longrightarrow \frac{4a^2}{\pi}. \quad (70)$$

Proof. The rescaled support function is

$$\tilde{h}_q(\varphi) = a(1-q) \sum_{n \geq 1} q^{n-1} |\cos(n\theta + \varphi_0 - \varphi)|.$$

For irrational rotation, insert the absolutely convergent Fourier series of $|\cos|$. The constant coefficient contributes $2a/\pi$. Every nonconstant mode contains

$$\frac{(1-q)e^{ik\theta}}{1-qe^{ik\theta}} \longrightarrow 0.$$

Its modulus is at most one, so absolute summability gives convergence uniformly in φ . Uniform convergence of support functions is exactly Hausdorff convergence of convex bodies. In the rational case, the Abel means of a periodic sequence converge to its period average, uniformly in φ , yielding (69). Area is continuous under Hausdorff convergence, and the limiting disk has area $\pi(2a/\pi)^2$. $\square$

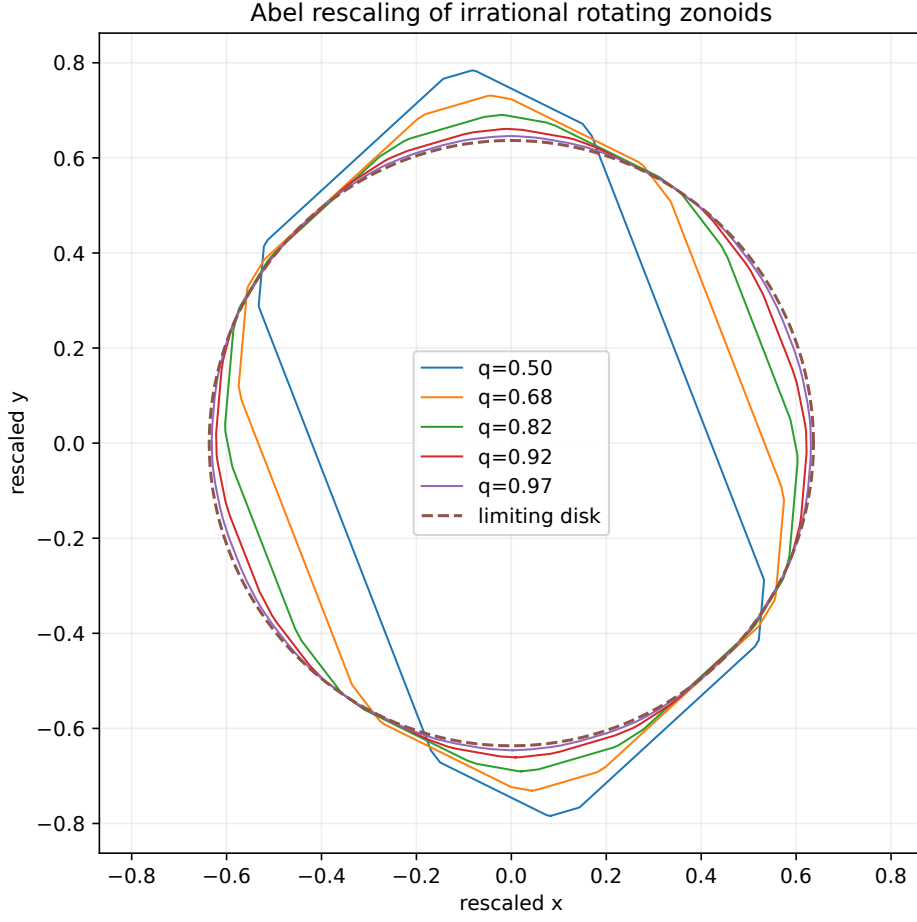


Figure 4: The rescaled supports $(1 - q)K_{q,\theta}/q$ approach the disk of radius $2/\pi$ for the golden-ratio rotation. Each finite plot uses enough generators that its omitted Hausdorff tail is below 2×10^{-8} .

Remark 10.9 (Geometry remembers arithmetic). The normalized perimeter is the same in both cases, but the limiting shape remembers whether the rotation is periodic. Rational angles retain a finite directional alphabet and produce a polygon. Irrational angles equidistribute the atomic curvature measure (59) and produce a disk. At finite q , however, the irrational support still has pure-point curvature with dense atoms. Smooth shape appears only after Abel rescaling and passage to the limit.

10.5 Cyclotomic formulas for projection cumulants

Let w make angle φ and put $\delta = \varphi_0 - \varphi$. Then $\langle w, q^n R_{n\theta} v \rangle = a \|w\| q^n \cos(n\theta + \delta)$. Let $Q = q^{2m}$.

Proposition 10.10 (Finite resolvent formula). *[proved here]* For $m \geq 1$,

$$\begin{aligned} \kappa_{2m}(\langle w, X_{q,\theta} \rangle) = c_{2m}(a \|w\|)^{2m} & \left[2^{-2m} \binom{2m}{m} \frac{Q}{1-Q} \right. \\ & \left. + 2^{1-2m} \sum_{\ell=1}^m \binom{2m}{m-\ell} \operatorname{Re} \left(e^{2i\ell\delta} \frac{Q e^{2i\ell\theta}}{1 - Q e^{2i\ell\theta}} \right) \right]. \end{aligned} \quad (71)$$

Thus every fixed-order projection cumulant is a finite sum of geometric or cyclotomic resolvents, even when the full density has no elementary expression.

Proof. Use

$$\cos^{2m} x = 2^{-2m} \binom{2m}{m} + 2^{1-2m} \sum_{\ell=1}^m \binom{2m}{m-\ell} \cos(2\ell x)$$

and sum each geometric Fourier mode over $n \geq 1$. $\square$

Example 10.11 (Quarter-turn factorization). If $v = e_1$ and $\theta = \pi/2$, the even and odd digits occupy orthogonal axes:

$$\begin{aligned} X_1 &= \sum_{k \geq 1} (-1)^k q^{2k} U_{2k}, \\ X_2 &= \sum_{k \geq 1} (-1)^{k-1} q^{2k-1} U_{2k-1}. \end{aligned}$$

The coordinates are independent scalar geometric-uniform laws with base q^2 and different scales. The support is a rectangle with half-widths $q^2/(1 - q^2)$ and $q/(1 - q^2)$, and (54) reduces to its area $4q^3/(1 - q^2)^2$.

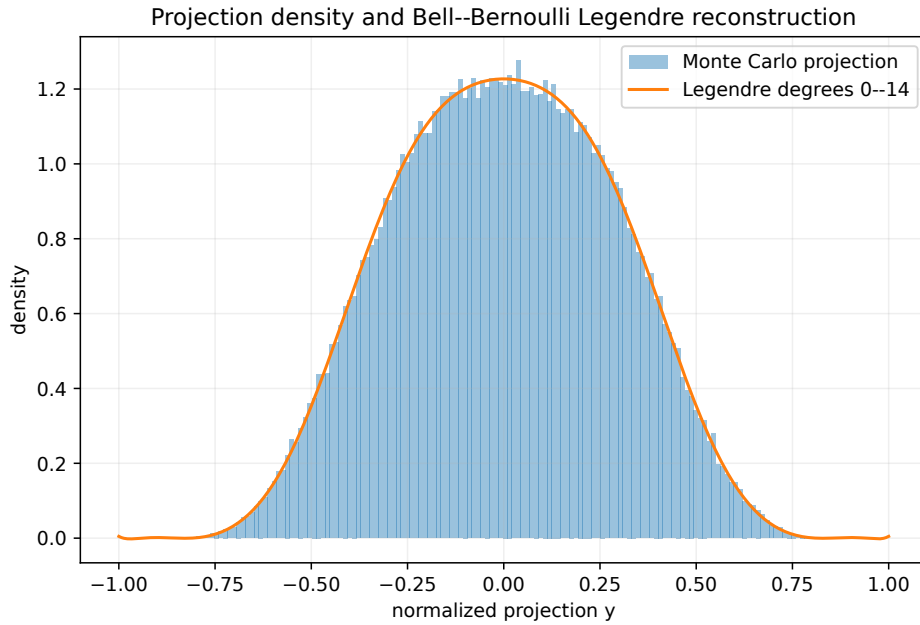


Figure 5: A degree-14 Legendre reconstruction of one normalized projection, with coefficients computed from (71), Bell moments, and (43). The histogram is independent Monte Carlo data.

11 Directional caps, generalized inverse Fabius functions, and Lambert scales

For a direction w , let $h = h_K(w)$ and define the deficit from the supporting hyperplane

$$D_w = h - \langle w, X \rangle \geq 0. \quad (72)$$

Its distribution function $C_w(x) = \mathbb{P}(D_w \leq x)$ is a directional analogue of a Fabius endpoint germ, and C_w^{-1} is a generalized inverse Fabius quantile.

Theorem 11.1 (Exact cap product). *[proved here] Put $a_{n,j} = |\langle w, B^n v_j \rangle|$. Then*

$$D_w \stackrel{d}{=} 2 \sum_{n,j} a_{n,j} V_{n,j}, \quad V_{n,j} \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1], \quad (73)$$

and its Laplace transform is

$$\mathcal{L}_w(s) := \mathbb{E} e^{-s D_w} = \prod_{n,j: a_{n,j} > 0} \frac{1 - e^{-2s a_{n,j}}}{2s a_{n,j}} \quad (s \geq 0). \quad (74)$$

Proof. For one digit with scalar projection coefficient b , $|b| - bU$ has the same law as $2|b|V$ with V uniform on $[0, 1]$. Independence then gives the sum and product. $\square$

Corollary 11.2 (Rational rotations reduce to finitely many scalar endpoint products). *[proved here] In the rotating planar family, suppose θ/π is rational and let p be a period of $c_n = |\cos(n\theta + \delta)|$. Then*

$$\mathcal{L}_w(s) = \prod_{\substack{1 \leq r \leq p \\ c_r > 0}} \prod_{k=0}^{\infty} \frac{1 - \exp(-2sa \|w\| c_r q^{r+kp})}{2sa \|w\| c_r q^{r+kp}}. \quad (75)$$

Thus a rationally rotating cap is a convolution of finitely many scalar geometric endpoint laws with common ratio q^p . The scalar saddle and lower Lambert–W machinery in the repository can be applied factor by factor.

Proof. Group the product (74) by residue class $n \bmod p$. $\square$

For irrational rotations, the amplitudes $q^n |\cos(n\theta + \delta)|$ have a geometric envelope but a nonperiodic multiplicative modulation. The logarithm of the large-argument part of (74) contains the rotation cocycle

$$\mathcal{C}_N(\theta, \delta) := \sum_{n=1}^N \log(2 |\cos(n\theta + \delta)|). \quad (76)$$

Its mean is zero, because $(2\pi)^{-1} \int_0^{2\pi} \log(2 |\cos x|) dx = 0$, but close returns to the zeros of cosine can create large negative excursions. This is the new obstruction to a literal periodic Lambert correction.

Conjecture 11.3 (Diophantine quasiperiodic endpoint law). *[conjecture] Suppose θ/π is of bounded continued-fraction type and the phase δ is nonresonant. Put $\lambda = \log(1/q)$. Then the cap Laplace transform has*

$$\log \mathcal{L}_w(s) = -\frac{(\log s)^2}{2\lambda} + O(\log s \log \log s) \quad (s \rightarrow \infty), \quad (77)$$

with a refinement whose nonuniversal term is an explicit smoothing of $\mathcal{C}_{\lfloor \log s / \lambda \rfloor}(\theta, \delta)$. After saddle-point inversion, $C_w(x)$ and $C_w^{-1}(y)$ have the same leading lower-Lambert skeleton as the scalar geometric law, but the scalar log-periodic carrier is replaced by a bounded or logarithmically growing quasiperiodic cocycle.

Conjecture 11.4 (Liouville resonant spikes). *[conjecture] For a dense set of Liouville angles and suitable phases, arbitrarily close returns to a cosine zero produce subsequences on which the correction to the quadratic term in (77) is larger than any fixed multiple of $\log \log s$. Consequently no bounded periodic or quasiperiodic carrier can give a uniform all-orders endpoint expansion.*

Remark 11.5. These conjectures are deliberately weaker than asserting a full expansion. The quadratic scale comes from the geometric envelope q^n ; the unresolved work is to control the singular Birkhoff sums (76) uniformly on the saddle window. The problem sits at an intersection of Fabius endpoint analysis, Sudler-type trigonometric products, Diophantine approximation, and lower-Lambert inversion.

12 Numerical experiments

The archive contains `matrix_fabius_experiments.py`, a deterministic, fully commented program using the seed 20260830. It regenerates all figures and writes machine-readable CSV tables. The principal example uses

$$q = 0.68, \quad \frac{\theta}{\pi} = \frac{\sqrt{5} - 1}{2}, \quad v = (1, 0).$$

The irrational angle is of bounded type and gives well-distributed directions.

12.1 Geometry checks

The 18-generator polygon and determinant formulas agree as follows:

Quantity	Value
polygon area	5.333 705 892 663 187 1
determinant-series area	5.333 705 892 663 188 0
absolute discrepancy	0.000 000 000 000 000 9
polygon perimeter	8.491 785 533 658 859 5
generator-sum perimeter	8.491 785 533 658 861 2
absolute discrepancy	0.000 000 000 000 001 8
infinite area	5.344 701 655 463 689 2
infinite perimeter	8.500 000 000 000 001 8

The area series was truncated only after a geometric upper bound on the omitted tail fell below 10^{-16} .

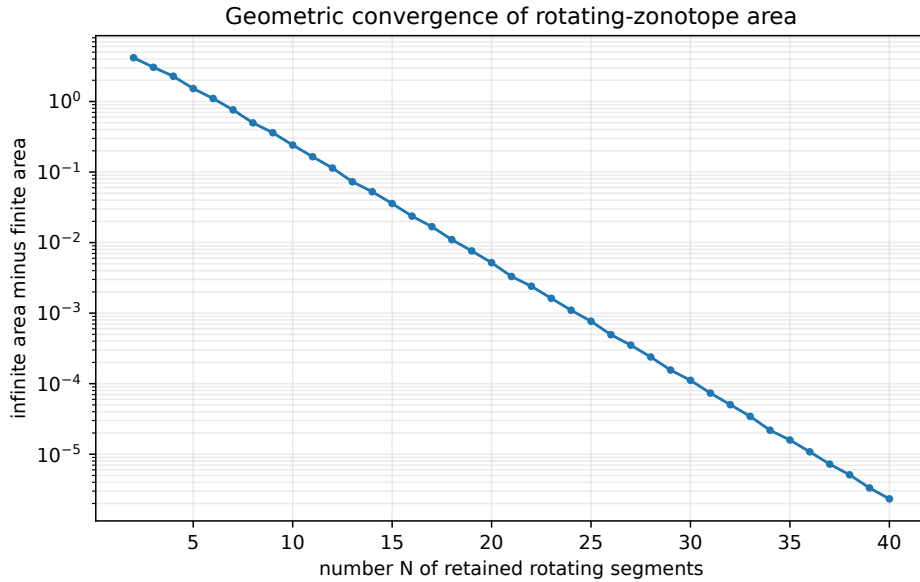


Figure 6: Convergence of finite support area to the exact infinite series (54).

12.2 Fourier, covariance, and Prouhet checks

For twelve random frequencies, the largest residual in the exact finite-product identity

$$\Phi_N(A^T \xi) = \text{sinc}_{\text{rad}} \langle \xi, v \rangle \Phi_{N-1}(\xi)$$

was 3.33×10^{-16} . For a six-generator directional Prouhet cube, all signed moments below degree six vanished with maximum residual 4.20×10^{-16} , and the first surviving degree matched (29) within 2.78×10^{-16} .

With 180,000 Monte Carlo samples, the sample covariance was

$$\begin{pmatrix} 0.09506634 & -0.02734876 \\ -0.02734876 & 0.19191718 \end{pmatrix},$$

while the exact infinite Lyapunov solution was

$$\begin{pmatrix} 0.09457932 & -0.02746051 \\ -0.02746051 & 0.19212703 \end{pmatrix}.$$

The largest entrywise discrepancy, 4.87×10^{-4} , is consistent with ordinary Monte Carlo error. The direct 28-generator covariance sum and the infinite Lyapunov solution agree to the displayed digits.

12.3 Reproduction command

From the extracted archive directory, run

```
python matrix_fabius_experiments.py --output-dir . --samples 180000
```

The script creates `figures/`, `data/`, and `results/`. It uses no network access and overwrites only its own generated outputs.

13 Formalization blueprint

The repository already contains reusable probability and infinite-product infrastructure for absolutely summable vector weights. The following sequence would minimize duplication in a Lean formalization.

13.1 Phase I: generic vector series and support

Define the matrix weights $g_{n,j} = A^{-n}v_j$ as a flattened summable family. Reuse the existing arbitrary-vector-weight law to obtain convergence, pushforward naturality, reflection, and support-as-range. Add a finite-dimensional lemma identifying the range with the infinite Minkowski sum and proving (3). The only matrix-specific analytic input is summability of powers of an expansive inverse.

13.2 Phase II: sinc products and smoothness

Instantiate the generalized Rvachev product with linear forms $z \mapsto \langle z, g_{n,j} \rangle$. Formalize the finite-frame lemma [Lemma 3.2](#), then the block-product decay estimate. The most delicate step is likely the passage from rapid Fourier decay to a compactly supported smooth density in several variables; Mathlib’s Fourier inversion API may require a Schwartz-space wrapper.

13.3 Phase III: distributions and derivative cubes

Prove (12) first for test functions, then use derivative and convolution commutation. The 2^M sum should be indexed by `Fin M -> Bool`; the sign is $(-1)^{\text{Fintype.card} \dots}$ or a direct parity character. The tensor Prouhet identity is best formalized from the factorized exponential polynomial before differentiating at zero.

13.4 Phase IV: tensor algebra

Represent order- k tensors as multilinear maps or symmetric powers according to available library support. A computationally lighter first tranche can state all results after contraction with arbitrary w ; polynomial extensionality then recovers the tensor identity. The operator geometric series

$$\sum_{n \geq 1} (A^{-1})^{\otimes kn} = (A^{\otimes k} - I)^{-1}$$

is the core reusable lemma.

13.5 Phase V: planar convex geometry

The perimeter and area formulas can initially be formalized for finite zonotopes, followed by monotone limits. The natural-boundary theorem requires only an absolutely summable Fourier series and a geometric-series calculation; it is analytically cleaner than a general natural-boundary criterion. The Abel shape theorem reduces to uniform convergence of support functions.

14 Open problems and further research

Open problem 14.1 (Optimal matrix Fourier decay). *Determine the sharp directional coefficient in $-\log |\Phi(tw)|/(\log t)^2$. For a scalar dilation it is governed by one logarithmic scale. For a matrix, the likely invariant is a pressure or joint Lyapunov quantity built from the orbit of w under A^{-T} and the finite frame V ; the block length L in (11) is only a robust upper bound.*

Open problem 14.2 (Smith-normal-form moment arithmetic). *For integral A, V , determine exact invariant-factor denominator formulas for joint moments. The matrices $A^{\otimes 2^m} - I$ should play the role occupied by $2^{2^m} - 1$ in the dyadic theory. It is natural to seek matrix q -binomial identities whose denominators are controlled by the Smith normal forms of these Kronecker powers.*

Open problem 14.3 (Zero-hyperplane spectral arithmetic). *The zero set of (6) is the union of hyperplanes*

$$\langle z, B^n v_j \rangle = k\pi, \quad k \in \mathbb{Z} \setminus \{0\}.$$

Classify multiplicities and intersections arithmetically. For rational matrices, coincidences reduce to lattice relations among rows of powers of A ; for rotations, they are controlled by cyclotomic or Diophantine relations. A multivariate Jensen formula may turn these arrangements into digit-counting identities analogous to the scalar zero divisor.

Open problem 14.4 (Matrix inverse Fabius geometry). *Develop uniform saddle asymptotics for the directional cap CDF C_w and its inverse as w crosses normal directions where one or more amplitudes $|\langle w, B^n v_j \rangle|$ vanish. Rational rotations suggest a finite collection of Lambert phases; irrational rotations suggest a stratified family of singular cocycles.*

Open problem 14.5 (Sharp plateau and strict-log-concavity phase diagram). *The non-strict log-concavity is settled by Theorem 3.5, while Proposition 10.3 gives a concrete open region of parameters with a flat core. Determine the exact set of (q, θ) for which the rotating density has a positive-area plateau, and decide whether it is strictly log-concave on $\text{int } K$ throughout the complementary region. The rational phase-locked cases reduce to products and finite convolutions of scalar geometric-uniform laws; irrational angles require controlling an infinite sequence of changing box-spline cells.*

Conjecture 14.6 (Curvature/density duality). *[conjecture] In the irrational rotating family, the rate at which the rescaled support approaches its limiting disk and the rate at which the standardized density*

approaches a Gaussian as $q \uparrow 1$ are controlled by the same small denominators $|1 - qe^{2ik\theta}|$. More precisely, suitable anisotropy norms of the support function and of the fourth-cumulant tensor should be comparable up to universal powers of $1 - q$ for bounded-type angles.

Open problem 14.7 (Multivariate polynomial reproduction). *The scalar repository contains exact polynomial synthesis by shifted up-atoms and Strang–Fix/Appell machinery. Determine the polynomial reproduction space of translates of $f_{A,V}$. The zero-hyperplane arrangement of Φ should replace scalar zero multiplicity, while the Appell tensors in Theorem 8.1 should give the correction operator. The answer may depend on an arithmetic compatibility between A , V , and the translation lattice.*

Open problem 14.8 (Noncommuting scale sequences). *Replace the single matrix power B^n by products $B_1 B_2 \cdots B_n$ from a finite noncommuting alphabet. The support remains a summable infinite zonotope and the derivative cube survives, but the Mahler equation becomes a cocycle and the cumulant resolvent becomes a transfer-operator series. This setting may link the Fabius–Rvachev system to joint spectral radius, matrix substitution systems, and automatic sequences beyond Thue–Morse.*

15 Conclusion

The scalar Fabius–Rvachev system is not an isolated special function. It is the rank-one member of a matrix-dilated family whose probability, harmonic analysis, convex geometry, arithmetic, and approximation theory remain exactly coupled. The matrix replacement preserves the decisive mechanisms:

- uniform digits produce sinc factors and Bernoulli cumulants;
- dilation produces a Mahler equation and operator resolvents;
- segment endpoints produce a Thue–Morse cube;
- finite prefixes are box splines and the limiting density is log-concave;
- infinite support is a zonoid, with an explicit plateau criterion;
- projection reduces tensor information to Bell moments and Legendre coefficients;
- directional endpoints recover inverse-Fabius and Lambert-type questions.

The rotating planar family shows that this extension is not merely formal. Arithmetic of the rotation angle is visible simultaneously in the support’s edge directions, its q -series area, analytic continuation, Abel shape limit, projection cumulants, and endpoint cocycle. Rational rotations give finite phase decompositions and rational functions; irrational rotations give dense edges, natural boundaries, and a circular Abel limit. These phenomena create a substantial frontier that is adjacent to, but not duplicative of, the repository’s existing scalar and base- b theory.

A Fourier and tensor conventions

The Fourier transform is

$$\widehat{f}_{\text{ang}}(-\xi) = \int_{\mathbb{R}^d} e^{i\langle \xi, x \rangle} f(x) \, dx, \quad f(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle \xi, x \rangle} \widehat{f}_{\text{ang}}(-\xi) \, d\xi.$$

Thus $\widehat{D_g f}_{\text{ang}}(-\xi) = -i \langle \xi, g \rangle \widehat{f}_{\text{ang}}(-\xi)$ and $\widehat{D_g^{2m} f}_{\text{ang}}(-\xi) = (-i)^m \langle \xi, g \rangle^{2m} \widehat{f}_{\text{ang}}(-\xi)$.

For vectors $x_1, \dots, x_k$, normalized symmetrization means

$$\text{Sym}(x_1 \otimes \cdots \otimes x_k) = \frac{1}{k!} \sum_{\sigma \in S_k} x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(k)}.$$

Its contraction with $z^{\otimes k}$ is $\prod_i \langle z, x_i \rangle$, which fixes the factor $k!$ in (29).

B A quantitative finite-prefix smoothness lemma

For completeness, here is the finite version implicit in [Theorem 5.2](#). Suppose the first L scales span and choose p consecutive spanning blocks. From each block, the frame argument selects a factor satisfying $|\langle \xi, g \rangle| \geq c_m \|\xi\|$. Hence

$$|\Phi_{pL}(\xi)| \leq C_p(1 + \|\xi\|)^{-p}.$$

If $p > d + k$, then $\|\xi\|^k \Phi_{pL} \in L^1$, so the corresponding finite box spline is C^k . Additional sinc factors cannot increase the real-axis modulus. Thus, for any prescribed derivative order, all sufficiently long finite prefixes possess that order uniformly. This argument avoids invoking a detailed combinatorial smoothness theorem for each box-spline direction multiset.

C Generated files and numerical ledger

The archive contains:

Path	Purpose
<code>matrix_dilated_fabius_rvachev.tex</code>	complete source of this report
<code>matrix_dilated_fabius_rvachev.pdf</code>	compiled report
<code>matrix_fabius_experiments.py</code>	commented deterministic experiment suite
<code>requirements.txt</code>	minimal Python dependency list
<code>README.md</code>	build and reproduction instructions
<code>CORPUS_AUDIT.md</code>	snapshot and novelty ledger
<code>figures/*.pdf, figures/*.png</code>	vector and raster figures
<code>data/area_convergence.csv</code>	finite/infinite area and perimeter comparison
<code>data/abel_residues.csv</code>	boundary Abel-residue tests
<code>data/abel_shape_limit.csv</code>	support-function distance to the limiting disk
<code>data/covariance_check.csv</code>	Monte Carlo, finite-sum, and Lyapunov covariance
<code>data/mahler_checks.csv</code>	finite matrix-Mahler residuals
<code>data/prouhet_checks.csv</code>	signed moment cancellations
<code>data/projection_legendre_coefficients.csv</code>	exact-formula numerical Legendre coefficients
<code>results/numerical_summary.txt</code>	human-readable audit trail

References

- [1] M. Anshelevich, Appell polynomials and their relatives, *International Mathematics Research Notices* 2004 (2004), no. 65, 3469–3531, doi: <https://doi.org/10.1155/S107379280413345X>.
- [2] J. Arias de Reyna, An infinitely differentiable function with compact support: definition and properties, English translation of *Rev. Real Acad. Cienc. Exact. Fis. Natur. Madrid* 76 (1982), 21–38, arXiv: <https://arxiv.org/abs/1702.05442>.

- [3] C. de Boor, K. Höllig, and S. Riemenschneider, *Box Splines*, Applied Mathematical Sciences 98, Springer, New York, 1993, doi: <https://doi.org/10.1007/978-1-4757-2244-4>.
- [4] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 5 (1966), 173–174.
- [5] E. Gover and N. Krikorian, Determinants and the volumes of parallelotopes and zonotopes, *Linear Algebra and its Applications* 433 (2010), 28–40, doi: <https://doi.org/10.1016/j.laa.2010.01.031>.
- [6] B. Jessen and A. Wintner, Distribution functions and the Riemann zeta function, *Transactions of the American Mathematical Society* 38 (1935), 48–88, doi: <https://doi.org/10.1090/S0002-9947-1935-1501802-5>.
- [7] P. McCullagh, Tensor notation and cumulants of polynomials, *Biometrika* 71 (1984), 461–476, doi: <https://doi.org/10.1093/biomet/71.3.461>.
- [8] A. Prékopa, On logarithmic concave measures and functions, *Acta Scientiarum Mathematicarum* 34 (1973), 335–343.
- [9] V. Yu. Protasov and T. I. Zaitseva, Self-affine 2-attractors and tiles, *Sbornik: Mathematics* 213 (2022), 794–830, doi: <https://doi.org/10.1070/SM9682>.
- [10] H. Weyl, Über die Gleichverteilung von Zahlen mod Eins, *Mathematische Annalen* 77 (1916), 313–352, doi: <https://doi.org/10.1007/BF01475864>.
- [11] V. Reshetnikov, ProveIt: Fabius function formalization and research documents, snapshot 90b2d3587833dea1bc4a6bf0c36d92f7200396e6, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction>.